

PIYUSH GARG

Delhi NCR • erpiyushgarg17@gmail.com • LinkedIn • GitHub • Portfolio

EDUCATION

Bachelor of Technology in AI & Data Science

Expected Graduation: May 2028

Akhilesh Das Gupta Institute of Professional Studies

- Relevant Coursework: Deep Learning, Data Structures & Algorithms, Calculus, Probability
- Current CGPA: 7.8/10.0

Central Board of Secondary Education (CBSE 12th)

Graduated: June 2024

Bal Bhavan International School

- Score: 75%

PROJECTS

Diffusion-Based Generative Modeling for Sparse Detector Data

- Developed a denoising diffusion probabilistic model (DDPM) from scratch using PyTorch.
- Built a generative pipeline including forward diffusion, reverse sampling, and timestep-conditioned denoising.
- Evaluated model performance by comparing statistical properties of generated detector events with real data using energy distribution analysis.

DIV2K Image Super Resolution using SRGAN

- Trained a Super Resolution GAN on the DIV2K dataset for high-quality image reconstruction.
- Used curriculum training strategy (L1 pretraining → adversarial training) for stable GAN convergence.
- Focused on perceptual quality improvement and color fidelity in reconstructed images.

ECG Anomaly Detection using Autoencoders

- Built an autoencoder-based anomaly detection system for ECG time-series signals.
- Used reconstruction error to identify abnormal cardiac patterns in an unsupervised setting.
- Implemented preprocessing pipeline for robust signal normalization and feature extraction.

Fashion-MNIST Image Generation using DCGAN

- Developed a Deep Convolutional GAN to generate synthetic fashion images.
- Designed convolutional generator and discriminator networks.
- Evaluated training stability and sample quality across epochs.

SKILLS

- Machine Learning:** PyTorch, TensorFlow, Scikit-Learn, Generative Models (GANs, Diffusion, Autoencoders)
- Programming:** Python, C++, C, JavaScript
- Scientific Computing:** NumPy, Pandas, Matplotlib
- Web:** HTML, CSS, React
- Languages:** English, Hindi